Feature-definition formatting proof

This proof contains technical definitions only; it is not the final manuscript.

For DB5, all 21089 saved feature rows were reconstructed exactly from the ten raw exercise files. The common filter function clips the requested upper cutoff to $\min(f_{\text{requested}}, 0.45 f_s - 1)$, giving 10–89 Hz at its sampling rate, although the original extraction report printed the requested 95 Hz. The filtering code and saved features are preserved; this correction changes reporting, not model inputs or outcomes. The band-pass design argument is four, giving eight poles before forward–backward application. GRABMyo's requested upper limit is below this clipping cap.

For one channel, let x_t, $t = 1, \ldots, T$, denote the filtered window, $\bar{x}$ its mean and $d_t = x_{t+1} - x_t$. The implemented amplitude features are

$$\text{MAV} = T^{-1} \sum_{t=1}^{T} |x_t|, \qquad \text{RMS} = \sqrt{T^{-1} \sum_{t=1}^{T} x_t^2},$$

$$\text{VAR} = T^{-1} \sum_{t=1}^{T} (x_t - \bar{x})^2, \qquad \text{WL} = \sum_{t=1}^{T-1} |d_t|,$$

$$\text{IEMG} = \sum_{t=1}^{T} |x_t|, \qquad \text{WAMP} = \sum_{t=1}^{T-1} \mathbf{1}\{|d_t| \geq \tau\}.$$

Variance uses divisor T. IEMG and waveform length are sample sums, without division by sampling rate. The count rules are

$$\text{ZC} = \sum_{t=1}^{T-1} \mathbf{1}\{x_t x_{t+1} < 0,\ |d_t| \geq \tau\},$$

$$\text{SSC} = \sum_{t=2}^{T-1} \mathbf{1}\{a_t b_t > 0,\ \max(|a_t|, |b_t|) \geq \tau\},$$

where $a_t = x_t - x_{t-1}$ and $b_t = x_t - x_{t+1}$. The thresholds are 10^{-2} mV for monopolar GRABMyo, 4×10^{-3} mV for derived bipolar signals and seven stored ADC units for DB5; they are not interchangeable amplitude units.

For spectral features, SciPy Welch estimation uses a periodic Hann window, segment length $\min(T, 256)$, default half-segment overlap, constant detrending and one-sided density scaling. Mean frequency is $\sum_\ell f_\ell P_\ell / (\sum_\ell P_\ell + 10^{-12})$. Median frequency is the first frequency bin whose cumulative power reaches half of that guarded total, clamped to the last bin if none reaches it; no frequency interpolation is used. The feature order within every channel is MAV, RMS, VAR, WL, ZC, SSC, IEMG, WAMP, MNF, MDF. This channel-major mapping is retained when selecting electrodes.